When Should an Instruction Filter Reject a Repeated Answer?

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CSC5051 Natural Language Processing — Assignment 2, Task 3

Abstract

An answer copied four times can still be correct. Should a data filter discard it or send it for review? We tested three prompts and two inspection-agent conditions on the same 32 GPT-4 Alpaca records, each paired with a deliberately damaged copy. Requiring full marks on four criteria kept 25–27 originals and admitted none of the 32 damaged copies under any method. Every method rejected empty, unrelated and echoed answers. Repetition accounted for all disagreements over the severity of an injected defect. A single zero-shot call per record produced as many outright rejections as the inspection agent (27/32), at an estimated CNY 0.111 rather than 0.210 for the test set. The agent retained two more originals.

1 What qualifies an example for training?

Instruction tuning uses an answer as a demonstration of how to carry out a request. An accurate but repetitive answer also demonstrates a style; a fluent answer can miss a constraint entirely. For each record, the judge assesses the answer against the instruction and optional input on four dimensions. *Instruction following* measures compliance with constraints; *correctness* measures support for factual or computational claims; *relevance and completeness* asks whether the response supplies what the user needs; and *clarity* covers readability and unnecessary repetition. Each score is 0 (fails), 1 (partly meets), 2 (meets), or null (cannot assess).

The judge keeps a record only when all four scores are 2, rejects it if any score is 0, and sends all other records for review. An empty optional input is acceptable. Only a valid *keep* enters the selected subset D' of the source records D. Scores of 1 and 0 therefore have the same immediate effect on admission, but only the former sends an otherwise acceptable record to a reviewer.

Design of the paired test. We sampled from the 52,002 English GPT-4 Alpaca records of Peng et al. (2023). To bound request length, eligible instructions were 20–400 characters, inputs at most 600, outputs 60–800, and the total at most 1,400 characters, with at least 95% ASCII. We excluded instructions containing terms such as "copy", "repeat" or "empty", then deduplicated instruction–input pairs. Seed 20261028 selected eight development bases and 32 disjoint test bases from 25,117 eligible records. The source version is pinned to commit `38b93b75`; the data carry a CC BY-NC 4.0 licence.

Each test base supplied one original and one constructed counterpart. In groups of eight, we emptied the output, substituted a donor reply, echoed the instruction and input, or repeated the original reply four times. Donors came from outside the selected bases and shared no filtered content words with the target instruction/input. Source replies have no human quality labels, so original retention measures selection. The controlled counterparts let us ask two observable questions: did the judge exclude the injected defect, and did it reject it outright?

A missing modal verb. Source 49960 asks for a rewrite using a modal verb. The original reply replaces "He should complete the project by the end of the month" with "It is necessary that he complete the project by the end of the month." Every judge flags the missing modal: zero-shot and stepwise score instruction following 0 (reject), while the other three score it 1 (review). All reject the unrelated winter poem in its damaged counterpart. Thus a retained original is not necessarily the right outcome for a pair.

2 How the judges make a decision

Each condition received the same three record fields and rubric. The system message treated record contents as data; identifiers and defect labels stayed outside the prompts and tool. The model returned four scores, a lowercase decision and a short reason. A parser checked the keys, score values and consistency with the decision rule. The conventional prompts varied the guidance preceding that contract:

Zero-shot. "Apply the supplied rubric directly." With no demonstrations, this measures how far the criteria and decision rule alone can guide the judge.

Few-shot. Two examples showed how a single failing dimension controls the decision (Brown et al., 2020). Given "Add the two integers. Respond with the number only" and input "2, 3", the answer "5" received four 2s and a keep. For "7", correctness alone fell to 0 and the decision was reject. These hand-written demonstrations were separate from the sampled data.

Stepwise. "Identify the instruction, its constraints, and the input context" precedes an assessment of each dimension, a check for contradictions and unsupported claims, and the final rule. This zero-shot chain-of-thought prompt (Kojima et al., 2022) directs attention to constraints before scoring. It uses no worked reasoning demonstrations (Wei et al., 2022) and collects a concise justification.

Inspection agent and control. The agent can `finish` with a scored assessment, `inspect` the record, or `reassess` a disputed criterion (Yao et al., 2023). Its prompt requests inspection for an empty output, a reply exceeding 120 words (including uncertain cases near that threshold), or apparent sentence or paragraph repetition. Other records can finish immediately. On inspection, Python returns empty fields, character and word counts, the fraction of distinct output words and the largest word frequency. The model sees the observation and its remaining call allowance before choosing its next action. Reassessment requests another reading without adding evidence. A three-call limit bounds the cost. The control uses the same agent prompt and limit but replaces the tool measurements with a message that the observation was withheld (Figure 1).

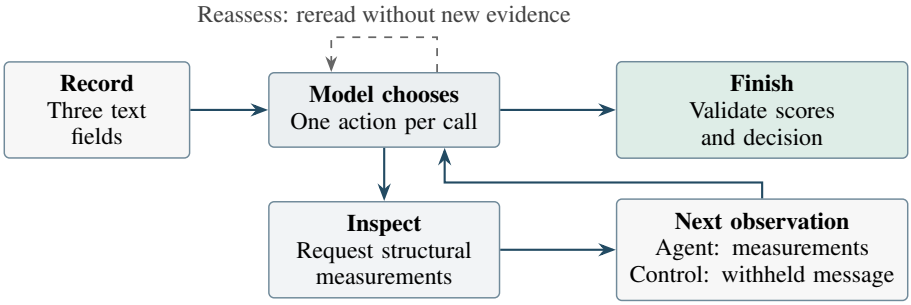


Figure 1: How feedback enters the next decision. Of 64 records per condition, the agent inspected 18 and the control inspected 17. The dashed reassessment branch was available but unused. Both had a three-call limit; neither used a third call.

Refining the prompts on development data. On four development pairs, three of 40 judgments failed because the stepwise prompt produced `Reject` instead of `reject`; all 16 agent judgments finished immediately. We spelled out the lowercase decision strings and added the inspection triggers above. The revised pilot on the same records produced 40 valid judgments; each agent condition inspected two records. We then froze the prompts before the held-out run.

Common model settings. The five methods processed the same 64 held-out records on 28 October 2026. Volcengine Ark returned `deepseek-v4-1-flash-260910`. Each request used temperature 0, `reasoning_effort=none` and a 768-token completion cap; three workers handled the run. Holding provider thinking fixed allows a comparison of prompting instructions.

API implementation. The Python request below shows the inference settings. The client reads the key locally, reserves the estimated maximum cost before sending, and saves the response and token usage under the run identifier. The supplement provides complete prompt templates and the client implementation.

```
payload = {"model": configured_model, "messages": messages,
        "temperature": 0, "reasoning_effort": "none",
        "max_completion_tokens": 768}
request = Request(
   url, method="POST", data=json.dumps(payload).encode(),
   headers={"Authorization": "Bearer " + api_key,
          "Content-Type": "application/json"})
with opener.open(request, timeout=120) as response:
   result = json.loads(response.read())
```

3 THE STRICT FILTER EXCLUDED EVERY CONSTRUCTED DEFECT

All five methods withheld every damaged answer. On the 24 empty, substituted and echoed answers, every method issued an outright rejection. The remaining eight replies contained the original content four times (Figure 2). Zero-shot rejected three and sent five for review; few-shot and stepwise each rejected one and sent seven for review. Both agent conditions rejected three, sent four for review and produced one malformed final JSON response. Each malformed response lacked a closing brace; both counted as invalid. Across all methods, 318 of 320 judgments were valid.

Table 1: Held-out filtering decisions. Each original group and each damaged group contains 32 records; no damaged record received a keep. A paired success requires keeping the original *and* rejecting its damaged counterpart.

	Original			Damaged			Paired
Method	Keep	Review	Reject	Reject	Review	Invalid	/32
Zero-shot	25	5	2	27	5	0	20
Few-shot	27	4	1	25	7	0	21
Stepwise	25	5	2	25	7	0	20
Inspection agent	27	4	1	27	4	1	23
Withheld feedback	27	4	1	27	4	1	22

Zero-shot and the agent both rejected 27 damaged replies (Table 1). The agent kept 27 originals versus 25 for zero-shot and produced the most pairs with both outcomes (23 versus 20). The modal-verb case in Section 1 illustrates why the paired count must be read alongside the original decisions.

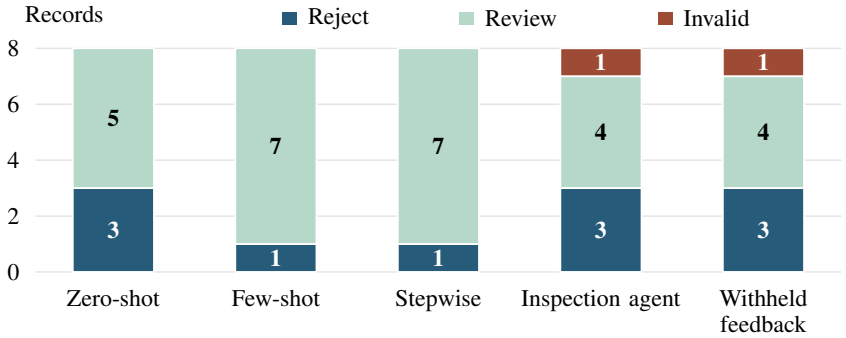


Figure 2: How each method handled the same eight answers repeated four times. Counts inside each bar distinguish rejection, referral for review, and invalid JSON. No method kept a repeated answer.

A repeated explanation: one answer, two severity judgments. In record 39766, a business closes a store after financial difficulties. The original reply connects the closure to insufficient revenue or high expenses, and every method keeps it. When the same paragraph appears four times, all three conventional prompts still award full marks for correctness and score clarity 1. Their decision is review. The agent asks to inspect and receives a word count of 272, a distinct-word fraction of 0.180, and a maximum word frequency of 24. Its final reason cites the length and repetition; it assigns clarity 0 and rejects. The withholding control also inspects, then scores clarity 1 and sends the record for review. Both routes keep the repeated answer out of training. Their difference is whether a person must resolve it.

What changes when the threshold moves? We reapplied two more permissive rules to the saved scores. Allowing one partial score (total at least 7, no dimension below 1) caused zero-shot to admit four additional originals *and five* repeated answers. The agent admitted four repeated answers with no increase in original retention. Across methods, all newly admitted diagnostics were repetitions. Table 2 shows the consequence of a policy change with the model judgments held fixed.

Table 2: Originals / damaged replies admitted out of 32 each, using the same judgments. Both relaxed rules require all four scores to be non-null.

Method	All four $= 2$	All ≥ 1, sum ≥ 7	All ≥ 1
Zero-shot	25 / 0	29 / 5	30 / 5
Few-shot	27 / 0	28 / 5	30 / 7
Stepwise	25 / 0	27 / 5	28 / 7
Inspection agent	27 / 0	27 / 4	29 / 4
Withheld feedback	27 / 0	27 / 4	29 / 4

4 What did inspection buy?

The agent requested structural measurements for 18 of 64 records; the withholding control requested them for 17. Neither selected reassessment or used a third call. Each skipped three outputs that exceeded the stated length trigger (123, 124 and 131 words). The prompt delegated the choice to the model, and these traces show where it followed or missed the instruction.

The two agent conditions differed on three diagnostic decisions and one paired success, yet their aggregate decision counts were identical. Each condition chose its own actions, so a single pair of traces cannot isolate the effect of its observation.

Table 3: Resource use for the same 64 records, including invalid judgments. Mean and median seconds sum request latencies within each record. Estimated CNY uses measured token counts at peak list prices (2 per million input, 8 per million output), without cache discounts (Volcengine Ark, 2026).

Method	Calls	Input tokens	Output tokens	Mean s	Median s	Est. CNY
Zero-shot	64	34,519	5,259	1.40	1.34	0.1111
Few-shot	64	47,191	4,470	1.28	1.25	0.1301
Stepwise	64	37,655	5,419	1.39	1.34	0.1187
Inspection agent	82	78,286	6,655	1.80	1.56	0.2098
Withheld feedback	81	76,100	6,560	1.77	1.50	0.2047

Zero-shot used 64 calls and an estimated CNY 0.1111. The agent's 18 follow-up calls brought it to 82 calls and CNY 0.2098, an 88.8% increase; mean per-record latency rose from 1.40 to 1.80 seconds. The 355 held-out calls together cost an estimated CNY 0.774406. Including both pilots and the initial connectivity check, the estimate was CNY 0.948306. These figures price observed token use; billing and network latencies can vary.

Practical choice. For screening similar short records under this rubric, zero-shot is the economical starting point: one call per record excluded all constructed defects, and its outright rejection count matched the agent's. Inspection supplies an auditable measurement for a severity judgment. To choose a threshold for a training corpus, the next useful evidence would be human severity labels on naturally occurring repetition and factual errors. As with other LLM judges (Zheng et al., 2023), agreement among prompts from one model cannot establish human agreement. This paired test measures filtering decisions on short records; downstream training gains require a training evaluation.

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REFERENCES

Tom B. Brown et al. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020. URL https://arxiv.org/abs/2005.14165.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. *arXiv preprint arXiv:2205.11916*, 2022. URL https://arxiv.org/abs/2205.11916.

Baolin Peng, Chunyuan Li, Pengcheng He, Michel Galley, and Jianfeng Gao. Instruction tuning with gpt-4. *arXiv preprint arXiv:2304.03277*, 2023. URL https://arxiv.org/abs/2304.03277.

Volcengine Ark. Model pricing. https://console.volcengine.com/ark/region:cn-beijing/docs/model-pricing, 2026. Accessed 28 October 2026. Conservative peak list prices used; actual billing may differ.

Jason Wei et al. Chain-of-thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*, 2022. URL https://arxiv.org/abs/2201.11903.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.03629.

Lianmin Zheng et al. Judging LLM-as-a-Judge with MT-Bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, 2023. URL https://arxiv.org/abs/2306.05685.